

Islam I. Abdulaal

Integrated Photonics Researcher

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RESEARCH INTERESTS

Integrated nanophotonics, quantum photonic sources, nonlinear on-chip devices, physics-informed inverse design, and wave-equation-constrained neural networks for photonic device optimization.

EDUCATION

B.Sc. in Electronics and Communications Engineering Aug 2021 – Aug 2026
Alexandria University, Faculty of Engineering Onsite

RESEARCH EXPERIENCE

Research Intern Sep 2025 – Present
*NanoPhoto Lab, IMRE, A*STAR* Remote

Research Intern Jul 2025 – Sep 2025
Advanced Labs Program, Egypt Scholars Inc. Remote

Research Intern Jul 2023 – Aug 2025
OPST Group, SmartCI Research Center, Alexandria University Hybrid

SELECTED PUBLICATIONS

*Citation metrics based on Google Scholar (h-index: 2, citations: 11), July 2026.

Journal Articles

[1] **Abdulaal, I. I.**, Abdelrahman W. A. Elsayed, Omar A. M. Abdelraouf, “Recent advances in terahertz-based bound states in the continuum metasurfaces for 6G communications and MedTech,” *Journal of Optics*, 2026. Invited Paper
DOI: [10.1088/2040-8986/ae8605](https://doi.org/10.1088/2040-8986/ae8605)

Preprints

[1] Amir H. M. Labeab, Basmala Sallam, Abdelrahman W. Elsayed, **Abdulaal, I. I.**, Amany M. Kamal, Omar A. M. Abdelraouf, “Beyond Data-Driven: How Physics-Informed Neural Networks are Reshaping Multi-Physics Design and Discovery,” *Preprint*, 2026.
DOI: [10.48550/ARXIV.2606.21945](https://doi.org/10.48550/ARXIV.2606.21945)

HONORS, AWARDS, AND RESEARCH FUNDING

Dean's Honors for Academic Excellence 2021 -- 2025
Issuer: Alexandria University, Faculty of Engineering

ICMTC Artificial Intelligence Contest (AIC-2) -- 4th Place 2024
Issuer: Military Technical College

Huawei ICT Skills Competition -- National Finalist (AI Track) 2023
Issuer: Huawei Technologies -- MENA Region

NASA Space Apps Challenge -- Global Nominee 2021
Issuer: NASA

LEADERSHIP EXPERIENCE AND ACADEMIC SERVICE

Board Member Jul 2026 -- Present
Optica Egypt Local Section Hybrid

Chairman Feb 2025 -- Oct 2025
IEEE Solid-State Circuits Society (SSCS), Alexandria University Student Branch Chapter Hybrid

General Coordinator Aug 2021 -- Sep 2023
Education Clinic (USA-Registered NGO) Remote

TEACHING AND MENTORING

Tutor / Mentor, Optica Egypt

Jul 2026 – Present

PROFESSIONAL MEMBERSHIPS

Optica (formerly OSA)

Jul 2026 – Present

Student Member.

Institute of Electrical and Electronics Engineers (IEEE)

2025 – 2026

Student Member (ID: 101099759).

REFERENCES

Dr. Omar Abdelraouf

Research Scientist

Institute of Materials Research and Engineering (IMRE),

A*STAR, Singapore

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Prof. Ishac Kandas

Associate Professor

Department of Engineering Physics and Mathematics,

Alexandria University

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